TUTORIAL 6:

① Find the unit vector normal to the surface $x^3+y^3+z^3+3xyz=3$ at indicated point $(1,2,-1)$.

Soln: Let $\phi = x^3+y^3+z^3+3xyz-3$ be the surface.

$\therefore$ Normal vector $\Rightarrow \bar{N} = \nabla\phi$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$\nabla\phi = (3x^2+3yz)\hat{i} + (3y^2+3xz)\hat{j} + (3z^2+3xy)\hat{k}$$

$$\nabla\phi = 3\left[(x^2+yz)\hat{i} + (y^2+xz)\hat{j} + (z^2+xy)\hat{k}\right]$$

At point $(1,2,-1)$, $\nabla\phi = 3\left[(1-2)\hat{i} + (4-1)\hat{j} + (1+2)\hat{k}\right]$

$$\nabla\phi = \bar{N} = 3(-\hat{i}+3\hat{j}+3\hat{k})$$

$$|\bar{N}| = \sqrt{9+81+81} = 3\sqrt{19}$$

$$\hat{N} = \frac{\bar{N}}{|\bar{N}|} = \frac{\cancel{3}(-\hat{i}+3\hat{j}+3\hat{k})}{\cancel{3}\sqrt{19}}$$

$$\therefore \boxed{\hat{N} = \frac{1}{\sqrt{19}}(-\hat{i}+3\hat{j}+3\hat{k})}$$

② Find the unit tangent & unit normal vector to the curve $x=2t$, $y=t^2$, $z=t^2-1$ at $t=0$.

Sol: Let $\bar{r} = x\hat{i}+y\hat{j}+z\hat{k}$ be the position vector of point (x,y,z)

$$\bar{r} = (2t)\hat{i} + t^2\hat{j} + (t^2-1)\hat{k}$$

We know that, $\bar{T} = \frac{d\bar{r}}{dt} = 2\hat{i} + 2t\hat{j} + 2t\hat{k}$

At $t=0$, $\bar{T} = 2\hat{i}$

$$|\bar{T}| = \sqrt{4+4t^2+4t^2} = \sqrt{4+8t^2} = 2\sqrt{1+2t^2}$$

At $t=0$, $|\bar{T}| = 2 = \left|\frac{d\bar{r}}{dt}\right|$

Unit tangent vector $\hat{T} = \frac{\bar{T}}{|\bar{T}|} = \frac{\hat{i}+t\hat{j}+t\hat{k}}{\sqrt{1+2t^2}}$

~~At point t=2~~

At $t=0$,

$$\boxed{\hat{T} = \hat{i}}$$

Also, $\bar{N} = \left(\dfrac{\frac{d\hat{T}}{dt}}{\left|\frac{d\bar{r}}{dt}\right|}\right)$

$$\therefore \frac{d\hat{T}}{dt} = \frac{\sqrt{1+2t^2}\,(\hat{j}+\hat{k}) - (\hat{i}+t\hat{j}+t\hat{k})\frac{4t}{2\sqrt{1+2t^2}}}{1+2t^2}$$

At $t=0 \Rightarrow \dfrac{d\hat{T}}{dt} = \hat{j}+\hat{k}$

$$\therefore \bar{N} = \frac{d\hat{T}/dt}{|d\bar{r}/dt|} = \frac{\hat{j}+\hat{k}}{2}$$

$$|\bar{N}| = \frac{1}{\sqrt{2}}$$

unit normal vector $\boxed{\hat{N} = \dfrac{\bar{N}}{|\bar{N}|} = \sqrt{2}(\hat{j}+\hat{k})}$

(3) A particle moves along the curve $x = e^{-2t}$, $y = 3\cos 5t$, $z = 5\sin 2t$ where t is time. Find the components of velocity and acceleration at $t=0$.

Soln: Let $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of point (x,y,z)

$$\therefore \bar{r} = e^{-2t}\hat{i} + 3\cos 5t\,\hat{j} + 5\sin 2t\,\hat{k}$$

velocity $\bar{v} = \dfrac{d\bar{r}}{dt} = -2e^{-2t}\hat{i} - \dfrac{3\times 5\sin 5t}{\cancel{5}}\hat{j} + 10\cos 2t\,\hat{k}$

acceleration $\bar{a} = \dfrac{d\bar{v}}{dt} = +4e^{-2t}\hat{i} - 75\cos 5t\,\hat{j} - 20\sin 2t\,\hat{k}$

at $t=0$

$\bar{v} = -2e^{0}\hat{i} - 15\sin 0\,\hat{j} + 10\cos 0\,\hat{k} \Rightarrow \boxed{\bar{v} = -2\hat{i} + 10\hat{k}}$

$\bar{a} = +4e^{0}\hat{i} - 75\cos 0\,\hat{j} - 20\sin 0\,\hat{k} \Rightarrow \boxed{\bar{a} = 4\hat{i} - 75\hat{j}}$

(4) Find the angle between the normal to the surface $x^2yz=1$ at $(-1,1,1)$ and $(1,-1,-1)$.

Soln: Let $\phi = x^2yz-1$ be the required surface.

$\therefore$ Normal vector $\bar{N} = \nabla\phi$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$\nabla\phi = \bar{N} = (2xyz)\hat{i} + (x^2z)\hat{j} + (x^2y)\hat{k}$$

At $(-1,1,1)$, $\bar{N_1} = -2\hat{i}+\hat{j}+\hat{k}$

$(1,-1,-1)$, $\bar{N_2} = 2\hat{i}-\hat{j}-\hat{k}$

We know that,

$$\cos\theta = \frac{\bar{N_1}\cdot\bar{N_2}}{|\bar{N_1}|\cdot|\bar{N_2}|}$$

$|\bar{N_1}| = \sqrt{4+1+1} = \sqrt{6}$

$|\bar{N_2}| = \sqrt{4+1+1} = \sqrt{6}$

$$\therefore \cos\theta = \frac{(-2\hat{i}+\hat{j}+\hat{k})\cdot(2\hat{i}-\hat{j}-\hat{k})}{\sqrt{6}\sqrt{6}}$$

$$\cos\theta = \frac{-4-1-1}{6}$$

$$\cos\theta = -1$$

$$\theta = \cos^{-1}(-1)$$

$$\boxed{\theta = \pi}$$

(5) Find the angle between the tangents to $\bar{r} = t^2\hat{i}+2t\hat{j}-t^3\hat{k}$ at points $t=\pm 1$.

Soln: Given $\bar{r} = t^2\hat{i}+2t\hat{j}-t^3\hat{k}$

We know that $\bar{T} = \frac{d\bar{r}}{dt} = 2t\hat{i}+2\hat{j}-3t^2\hat{k}$

$\therefore$ at $t=1$, $\bar{T_1} = 2\hat{i}+2\hat{j}-3\hat{k}$

$t=-1$, $\bar{T_2} = -2\hat{i}+2\hat{j}-3\hat{k}$

Also $|\bar{T_1}| = |\bar{T_2}| = \sqrt{4+4+9} = \sqrt{17}$

We know that, $\cos\theta = \dfrac{\bar{T_1}\cdot\bar{T_2}}{|\bar{T_1}||\bar{T_2}|}$

$$\cos\theta = \frac{(2\hat{i}+2\hat{j}-3\hat{k}).(-2\hat{i}+2\hat{j}-3\hat{k})}{(\sqrt{17})(\sqrt{17})}$$

$$\cos\theta = \frac{-4+4+9}{17}$$

$$\cos\theta = 9/17$$

$$\boxed{\theta = \cos^{-1}(9/17)}$$